

Jake Doherty

Austin, TX

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Education

University of Texas at Austin

2022–Present

PhD Candidate, Physics

University of South Florida

May 2022

B.S. Physics, B.A. Mathematics

Technical Skills

Programming: Python, C++, C, Fortran, Bash

Scientific Computing: NumPy, SciPy, Matplotlib, Linux/Unix, Slurm, Git

Methods: Numerical integration of differential equations, Monte Carlo methods, Statistical modeling, Machine learning algorithms

High Performance Computing: Parallel and cluster computing, Large-scale numerical simulations

Research Experience

Graduate Researcher, University of Texas at Austin

Aug 2022–Present

Advisor: Pablo Laguna

- Design and implement numerical algorithms for solving coupled nonlinear partial differential equations using C++ and Fortran.
- Run large-scale simulations on HPC clusters using distributed computing resources.
- Analyze very large simulation datasets and build automated post-processing pipelines using Bash scripts and Python.

Research Assistant, University of South Florida

May 2020–Aug 2022

Advisor: Inna Ponomareva

- Modeled ferroelectric and antiferroelectric materials using computational physics methods.
- Performed numerical analysis of polarization, phase transitions, and electronic structure.

Selected Projects

Head-on Collisions of Boson Stars with Bowen–York Type Initial Data

- Developed a mathematical algorithm to construct constraint satisfying initial data for subsequent numerical evolution, and implemented said algorithm via modification of several C++ libraries.
- Implemented shooting method for solving coupled nonlinear PDEs
- Constructed piecewise cubic Hermite interpolating polynomials to interpolate among solutions.

Black Hole Growth in Scalar Field Cosmology

- Performed large scale numerical integration of PDEs on HPC clusters to study black hole spacetimes.
- Implemented numerical algorithms for manipulation of large datasets, including interpolation and plotting.

Publications

arXiv:2607.09494 [gr-qc] (2026)

Head-on Collisions of Boson Stars with Bowen–York Type Initial Data

Physical Review D 112, 044029 (2025)

Growth of a Black Hole in a Scalar Field Cosmology

Journal of Physical Chemistry C 125(29), 16296–16303 (2021)

Tunability of Structure, Polarization, and Band Gap of High- T_c Organic-Inorganic Ferroelectrics by Hydrostatic Pressure

Journal of Applied Physics 132, 034101 (2022)

Negative Capacitance Regime in Antiferroelectric PbZrO_3

Leadership & Teaching

Teaching Assistant, University of Texas at Austin

2022–Present

Directed Reading Program Mentor, University of Texas at Austin

2022–Present

Teaching Assistant, University of South Florida

2021–2022

Awards

USF Dean's List of Scholars (2021)

USF Director's Award (2018)

Florida Academic Scholars Award (2018)